

REAL -TIME-CHAT-APP

Overview -

This is a **real-time chat application** where users can:

- Signup / Login
 - Send & receive messages instantly
 - See online users
 - Chat in real-time using socket.io
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Simple Definition

A real-time chat app allows users to communicate instantly using Socket.io instead of traditional request-response systems.



Tech Stack



Frontend

- React.js
- Tailwind CSS
- Zustand (state management)
- daisyui



Backend

- Node.js
- Express.js
- MongoDB



Real-Time

- Socket.io
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Flow (Simple)

1. User logs in
2. Frontend sends request → Backend API
3. Backend verifies user (JWT)

4. Socket connection is created
 5. When user sends message:
 - Message → Backend
 - Backend → Socket emits
 - Receiver gets message instantly
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Real-Time Messaging Flow

User A → Server → User B ↓ Socket.io

✓ No refresh needed ✓ Instant updates



Folder Structure (Simple)

```
frontend/  
src/  
  components/  
  hooks/  
  lib/  
  pages/  
  store/
```

```
backend/  
  controllers/  
  routes/  
  models/  
  socket/
```



Features

-  Authentication (Login/Signup)
 -  Real-time chat
 -  Online users
 -  Multi-tab sync
 -  Responsive UI
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Deployment

- Frontend → Netlify pingchatify.netlify.app/

- Backend → Render chatify-nesy.onrender.com
 - Database → MongoDB Atlas
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Challenges Faced

- Cross-domain cookie issue
 - Real-time socket sync
 - Deployment differences (local vs production) // first it work proper in locally but not when i deploy this
 - Auth flow mismatch
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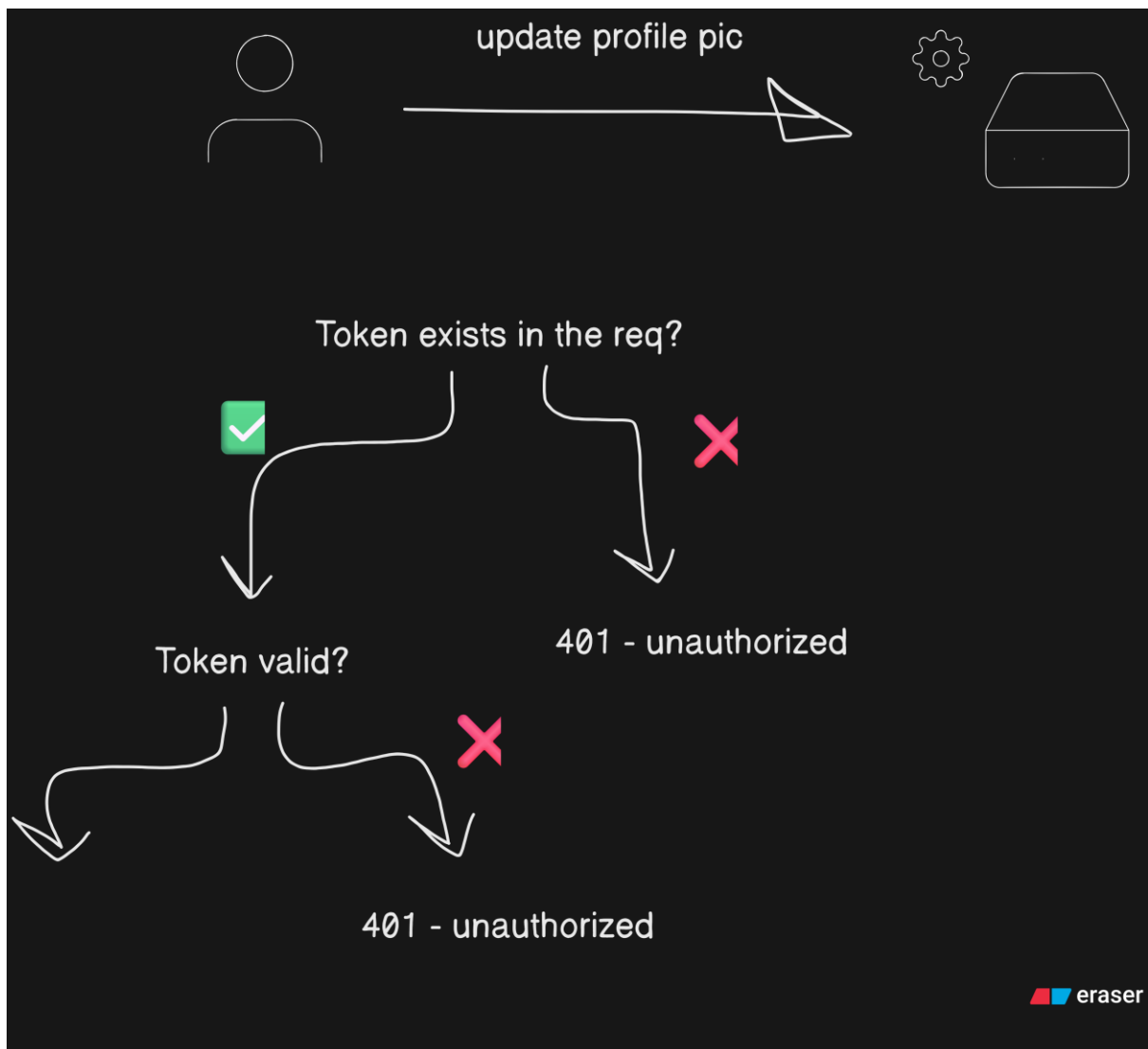
Solution

- Switched from cookies → token-based auth
 - Fixed axios interceptor
 - Fixed socket connection handling
 - Fixed Mis-matched ENV variable
-

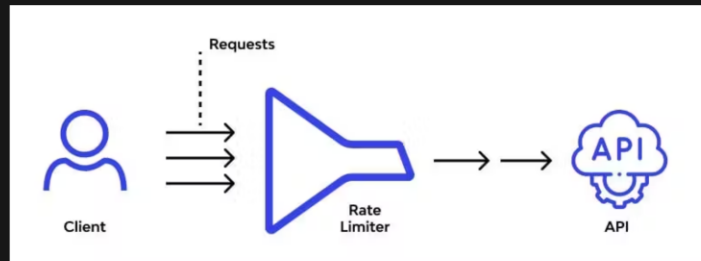
What I Learned

- Real-time systems using Socket.io
- Production debugging (CORS, auth, deploy issues)
- State management using Zustand
- Full-stack integration

Auth Workflow



RATE LIMITING



👉 **Rate limiting** is a way to control how often someone can do something on a website or app like how many times they can refresh a page, make a request to an API, or try to log in.

👉 Only 100 requests per user every 2 minutes

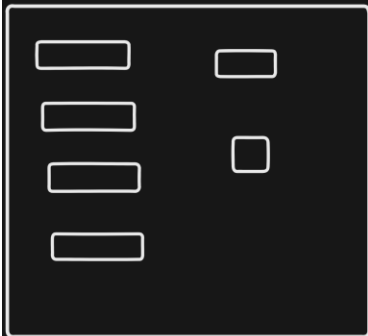
RATE LIMITING HELPS WITH

👉 Preventing abuse (e.g., stopping someone from making 1000 login attempts in a minute)

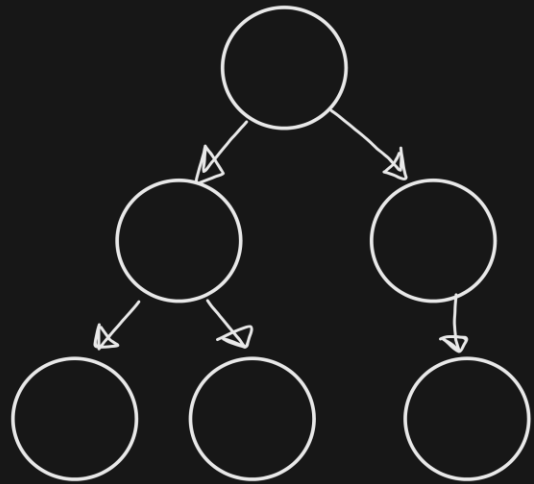
👉 Protecting servers from getting overwhelmed

Status Code? 👉 **429 Too Many Requests**

zustand store



App.jsx



my-chat-app.com



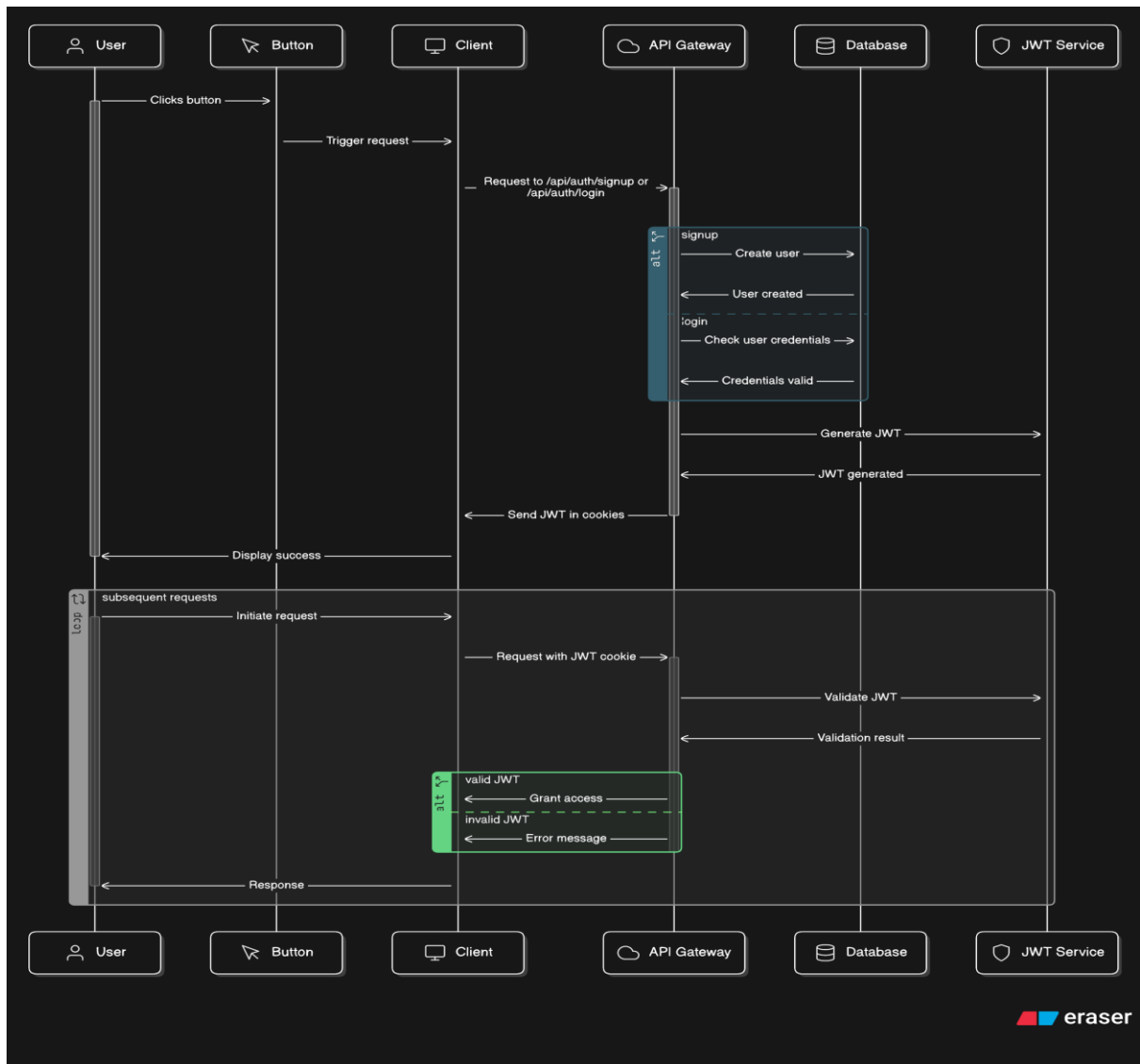
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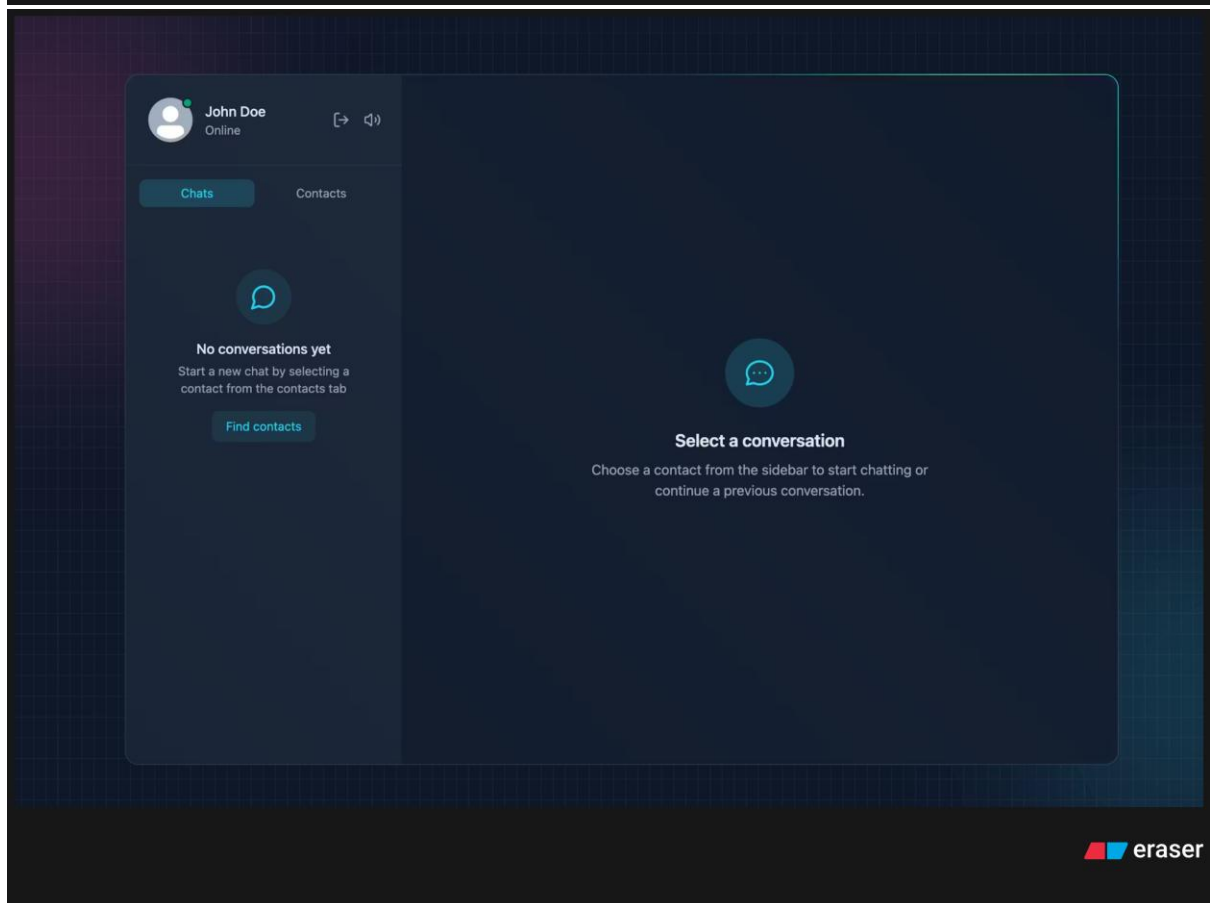
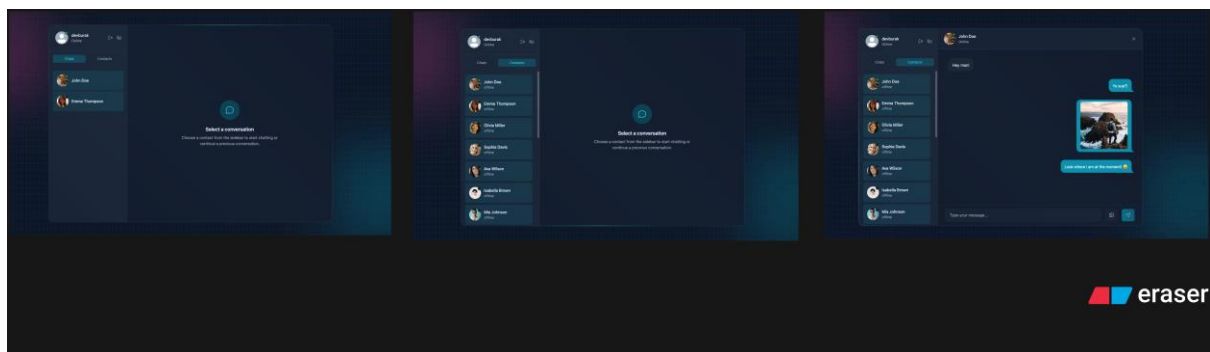
api

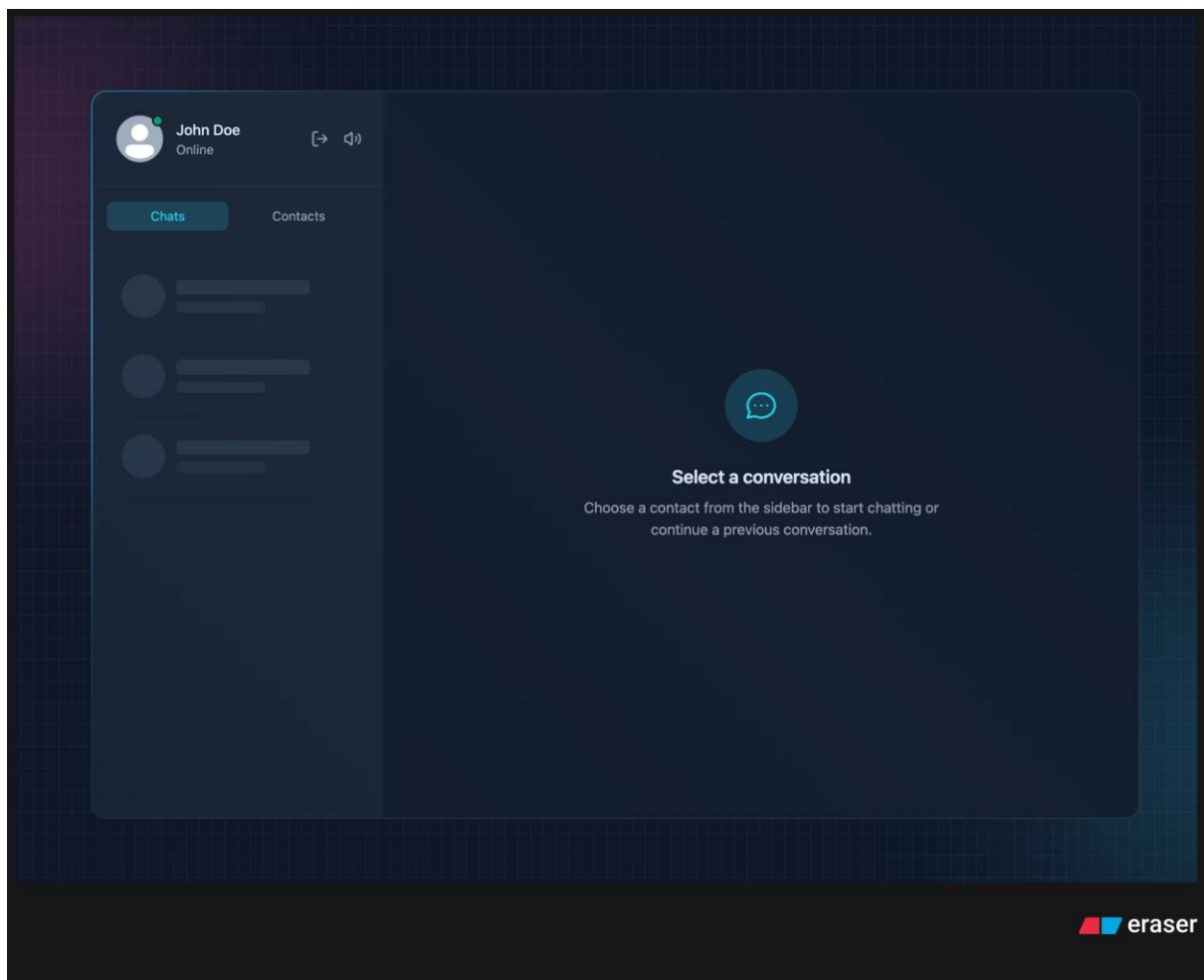


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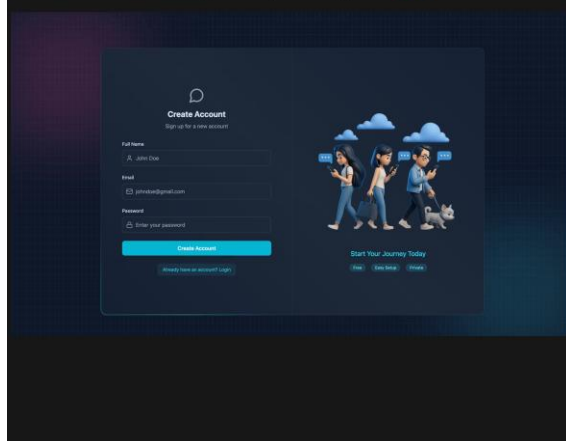
static assets







Sign Up Page



Login Page

